

Co-Training with Validation: A Generic Framework for Semi-Supervised Relation Extraction*

Shun Zhang[†]
Beijing Jiaotong University
Beijing, China
20120446@bjtu.edu.cn

Xiangkui Lu[†]
Beijing Jiaotong University
Beijing, China
21112021@bjtu.edu.cn

Jun Wu[‡]
Beijing Jiaotong University
Beijing, China
wuj@bjtu.edu.cn

ABSTRACT

In the scenarios of low-resource natural language applications, Semi-supervised Relation Extraction (SRE) plays a key role in mitigating the scarcity of labelled sentences by harnessing a large amount of unlabeled corpus. Current SRE methods are mainly designed based on the paradigm of Self-Training with Validation (STV), which employs two learners and each of them plays the single role of annotator or validator. However, such a single role setting under-utilizes the potential of learners in promoting new labelled instances from unlabeled corpus. In this paper, we propose a generic SRE paradigm, called Co-Training with Validation (CTV), for making full use of learners to benefit more from unlabeled corpus. In CTV, each learner alternately plays the roles of annotator and validator to generate and validate pseudo-labelled instances. Thus, more high-quality instances are exploited and two learners can be reinforced by each other during the learning process. Experimental results on two public datasets show that our CTV considerably outperforms the state-of-the-art SRE techniques, and works well with different kinds of learners for relation extraction.

CCS CONCEPTS

• **Artificial intelligence** → **Natural language processing**; • **Natural language processing** → **Information extraction**; • **Machine learning** → **Semi-supervised learning settings**.

KEYWORDS

Relation Extraction; Low Resource; Semi-supervised Learning; Natural Language Processing

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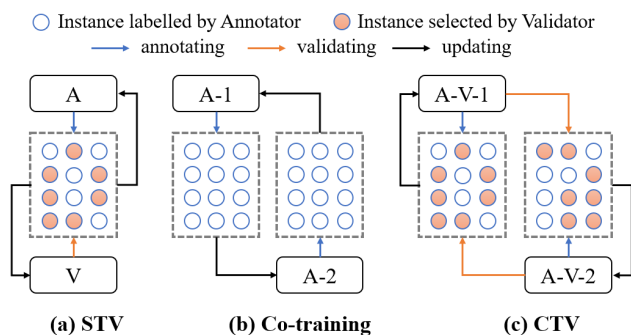


Figure 1: Comparison among (a) STV, (b) Co-training, and (c) CTV, assuming that the same learners' settings are used in the three paradigms. In each round of iterations, STV samples few yet high-quality pseudo-labelled instances from unlabeled corpus, while co-training utilizes larger number of noisy ones, and CTV strikes the balance between quality and quantity of the pseudo-labelled data.

1 INTRODUCTION

Relation Extraction (RE) aims to construct knowledge bases with relational facts from unstructured textual corpus, serving to a wide range of downstream natural language applications, such as information retrieval [3], question answering [5] and knowledge graph construction [11]. During the past years, a surge of efforts has been made to develop supervised RE models [13, 15, 17, 18, 20], which require a large number of labelled sentences for training. However, such a requirement on training data restricts its application coverage because large number of labelled sentences are fairly expensive to obtain, especially in some professional fields such as computer-aided diagnosis and drug discovery.

Although labelled sentences are scarce, unlabeled corpus are readily available and can be exploited in combination with labelled ones to perform RE tasks. Towards this direction, some Semi-supervised RE (SRE) methods are proposed [4, 6], which are designed based on the paradigm of Self-Training with Validation (STV). As illustrated by Figure 1 (a), STV employs two learners and each of them plays the single role of annotator or validator, i.e., those instances, firstly labelled by annotator and then checked

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[†] These authors contributed equally to this work.

[‡] Corresponding author.

by validator, are picked out for updating both of the learners. Despite enjoying many advantages, we argue that such a single role setting used by STV under-utilizes the potential of learners in comparison to the co-training paradigm [1]. In co-training (Figure 1 (b)), both of the learners are annotators, and thus around doubled pseudo-labelled instances can be used for updating learners. However, co-training suffers from the risk of learning from noisy labels introduced by pseudo-labelling.

In order to take the dual advantages of co-training and STV, we propose a novel SRE paradigm, called Co-Training with Validation (CTV), for pursuing both pseudo-labelling quantity and quality. Illustrated by Figure 1 (c), in CTV, each of the two learners plays the roles of both annotator and validator. By validating pseudo-labelled instances mutually, two learners are reinforced by each other during the iterative learning process. Specifically, our contributions are briefly summarized as:

- We propose a novel CTV paradigm for SRE tasks, which encourages two learners to switch the role between annotator and validator during the learning process. In contrast with STV, CTV can benefit more from unlabeled corpus. Different from co-training, CTV can generate higher quality pseudo-labelled instances towards stronger learning capability.
- The proposed CTV is a generic SRE framework that can employ different types of RE learners such as PRNN [18] and R-BERT [13].
- Experiments on two benchmarks show encouraging results in comparison to the state-of-the-art SRE techniques.

2 THE PROPOSED CTV PARADIGM

Let $(\mathbf{x}, \mathbf{y})$ be an instance that contains relation mention $\mathbf{x}$ and relation label $\mathbf{y}$, and $\mathcal{L} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^l$ as the labelled set, while $\mathcal{U} = \{\mathbf{x}_i\}_{i=l+1}^n$ denotes the unlabeled one, where l and n respectively denote the number of labelled instances and total amount of instances in relation mention corpus. In general, l is much smaller than n , which motivates us to devise the semi-supervised CTV paradigm that can jointly exploit $\mathcal{L}$ and $\mathcal{U}$ towards better generalization capability.

In our CTV framework, two RE learners (denoted as M_0 and M_1) are first initialized with $\mathcal{L}$, and then reinforced to each other by exploring and exploiting $\mathcal{U}$ iteratively. Particularly, each learner (annotator) first labels unlabeled sentences for its peer learner (validator); after that, the validator picks out high-quality instances from the noisy pseudo-labelled sentences and sends them back to the annotator for updating parameters. To better deploy base RE learners into CTV, we need to answer two additional questions that are not considered by the regular co-training paradigm [1].

2.1 How to train two RE learners with diversity?

As demonstrated in [12], it is not important whether the diversity is achieved by using two views or not. This provides a theoretical support to use co-training based on a single view. Fortunately, as deep neural networks (DNN) are highly nonconvex, even with the same network and optimization method, different initializations can also lead to different local optimal [2]. Considering this, we can take two DNN with the same architecture but different initializations as two diversified base RE learners. What's more, two base learners

Algorithm 1 The CTV paradigm

Input: labelled dataset $\mathcal{L}$, unlabeled dataset $\mathcal{U}$, maximum iterations T , sampling ratio τ , learners M_0 and M_1 with different initializations.

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1:  $\mathcal{L}_0 \leftarrow \mathcal{L}; \mathcal{U}_0 \leftarrow \mathcal{U}; M_0 \leftarrow \text{Train}(\mathcal{L}_0);$ 
2:  $\mathcal{L}_1 \leftarrow \mathcal{L}; \mathcal{U}_1 \leftarrow \mathcal{U}; M_1 \leftarrow \text{Train}(\mathcal{L}_1);$ 
3: while  $t \leq T$  do
4:    $t \leftarrow t + 1;$ 
5:   for  $s \in \{0, 1\}$  do
6:     Annotate  $\mathcal{U}_s$  by  $M_s$  to obtain:
        $\mathcal{P}_s = \{(\mathbf{x}, \hat{\mathbf{y}}^{(s)}) | \mathbf{x} \in \mathcal{U}_s, \hat{\mathbf{y}}^{(s)} = \text{Predict}(\mathbf{x}, M_s)\};$ 
7:     Sample  $\mathcal{P}_s$  by annotation confidence to obtain:
        $\mathcal{P}_s^- \leftarrow \text{Sample}(\mathcal{P}_s, \tau, \text{Max}(\hat{\mathbf{y}}^{(s)}), \text{descent});$ 
8:     Validate  $\mathcal{P}_s^-$  by  $M_{1-s}$  to obtain:
        $\tilde{\mathcal{P}}_s^- = \{(\mathbf{x}, \hat{\mathbf{y}}^{(1-s)}) | \mathbf{x} \in \mathcal{P}_s^-, \hat{\mathbf{y}}^{(1-s)} = \text{Predict}(\mathbf{x}, M_{1-s})\};$ 
9:     Sample  $\mathcal{P}_s^-$  by validation confidence to obtain:
        $\mathcal{L}_s^* \leftarrow \text{Sample}(\mathcal{P}_s^-, \tau, \text{CE}(\hat{\mathbf{y}}^{(s)}, \hat{\mathbf{y}}^{(1-s)}), \text{ascent});$ 
10:     $\mathcal{L}_s \leftarrow \mathcal{L}_s \cup \mathcal{L}_s^*; \mathcal{U}_s \leftarrow \mathcal{U}_s - \{\mathbf{x} | \mathbf{x} \in \mathcal{L}_s^*\};$ 
11:   end for
12:    $M_0 \leftarrow \text{Train}(\mathcal{L}_0); M_1 \leftarrow \text{Train}(\mathcal{L}_1);$ 
13: end while
```

Output: Stronger M_s ($s \in \{0, 1\}$) between two learners.

maintain their own training sets during the learning process. As two learners are diversified, the pseudo-labelled instances generated and selected by each learner for updating are not identical, which can further enlarge the diversity between them. Consequently, the diversity between M_0 and M_1 enables them to acquire different knowledge from unlabeled data, which is essential in promoting the cooperation between two RE learners.

2.2 How to exploit unlabeled data safely?

In order to make the proposed CTV paradigm robust against noisy pseudo labels, we devise two confidence assessment mechanisms. Concretely, a RE learner can be formulated as a predictive function: $\hat{\mathbf{y}}^{(s)} = M_s(\mathbf{x})$, where $s \in \{0, 1\}$ and $\hat{\mathbf{y}}^{(s)}$ is a probability distribution of $\mathbf{x}$ over all predefined relation categories, and the largest component $\hat{y}_j^{(s)}$ of $\hat{\mathbf{y}}^{(s)}$ can be regarded as the prediction confidence of M_s . Therefore, those pseudo-labelled instances with largest $\hat{y}_j^{(s)}$ from M_s are delivered to its peer learner M_{1-s} . Then the validator also makes its own prediction $\hat{\mathbf{y}}^{(1-s)} = M_{1-s}(\mathbf{x})$, and calculates the difference between $\hat{\mathbf{y}}^{(s)}$ and $\hat{\mathbf{y}}^{(1-s)}$ by $\Delta = \text{CE}(\hat{\mathbf{y}}^{(s)}, \hat{\mathbf{y}}^{(1-s)})$, where CE denotes the cross-entropy function. Thus, a pseudo-labelled instance $(\mathbf{x}, \hat{\mathbf{y}}^{(s)})$ is confident if Δ is small and it can be added into the training set of M_s , otherwise it will be discarded. Due to the diversity between M_s and M_{1-s} , their agreement on unlabeled data can be used to validate the *quality* of pseudo-labelled instances, which follows the spirit of manifold assumption that similar inputs result in similar outputs even though different learners are used.

2.3 Algorithmic framework

So far, we can put everything together to form the CTV for SRE. The algorithmic framework is shown in Algorithm 1, where $\text{Train}(\mathcal{L}_s)$ returns a learner trained with $\mathcal{L}_s$, $\text{Predict}(\mathbf{x}, M_s)$ returns predicted labels of $\mathbf{x}$ generated from M_s , $\text{Sample}(A, b, c, \text{'descent'/'ascent'})$ returns a set by sampling $b\%$ instances from set A according to the c

Table 1: Statistics of datasets, where ‘% Neg.’ denotes the percentage of negative instances with the label of ‘no_relation’.

Dataset	#Train	#Dev	#Test	#Relations	% Neg.
Semeval	7,199	800	2,715	19	17.60
Re-TACRED	58,465	19,548	13,418	40	71.57

values ranked in ‘descent/ascent’ order, $Max(y)$ returns the maximal element in vector y . The learning procedure stops when the maximum iterations T is reached or unlabeled data is used up. The final prediction is made by the stronger one between two learners.

3 EXPERIMENTS

In this section, we experimentally demonstrate the advantages of the proposed CTV paradigm.

3.1 Experimental setup

We conduct experiments on two public datasets and the statistics are showed in Table 1. **Semeval** and **Re-TACRED** are strikingly different in data scale, number of relations and the degree of imbalance between positive and negative instances. To better study the performance of CTV with different sizes of labelled set, we follow previous works [4, 6] to sample ratios of training set to form various labelled sets (5%, 10%, 30% for Semeval and 3%, 10%, 15% for Re-TACRED) and sample another 50% to form unlabeled sets. We adopt *micro-average Precision, Recall and F1-score* to evaluate the RE performance and we ignore the correct prediction for ‘no_relation’.

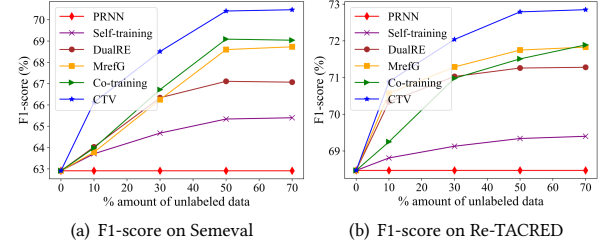
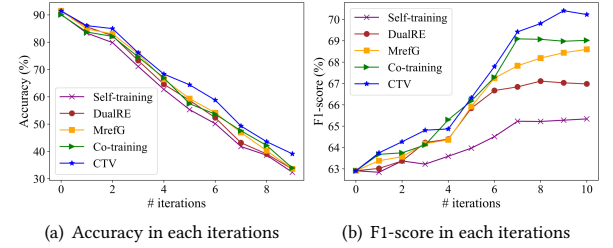
We compare our CTV with a supervised method (**PRNN** [18]) and four semi-supervised methods (**Self-training** [10], **DualRE** [6], **MrefG** [4] and **Co-training** [1]). We also apply a fully-supervised method (**PRNN-gold**) to give the upper-bound performance.

We initialize the word embeddings by the pre-trained GloVe-300d vectors and obtain POS tags and NER tags with Stanford CoreNLP [7]. We set the embeddings’ dimensions of position, POS tags and NER tags to 30. We utilize SGD as optimizer and set the learning rate to 1.0, the learning rate decay to 0.9, the batch size to 20 and the sampling ratio τ to 0.9.

3.2 Experimental results

3.2.1 Comparison with baselines. The performances of our CTV compared with baselines are shown in Table 2 and Table 3, where the best performance is boldfaced (excluding PRNN-gold). Each result is averaged over three independent repetitions and the standard deviation is reported. Some observations are as follows:

- Semi-supervised methods outperform supervised method PRNN in most cases, which reveals that unlabeled corpus are helpful for performance improvement of RE task.
- Among the five semi-supervised methods, self-training performs worse than others, especially when labelled data is very scarce. A major reason is that self-training uses a single learner, and does not apply any mechanism for labeling confidence validation.
- The two STV methods (DualRE and MrefG) achieve comparable performances with co-training. An explanation is that two STV solutions utilize less but higher quality pseudo-labelled instances, while co-training exploits more yet noisy

**Figure 2: Effect of the amount of unlabeled corpus.****Figure 3: Quality of pseudo-labelled instances and its influence on CTV with 10% labelled data on Semeval.**

ones. This observation demonstrates that both quality and quantity of pseudo-labelled instances are significant to improve the performance.

- It is impressive that our CTV always performs the best among four SRE methods and even fairly close to PRNN-gold in some cases, as both quantity and quality of pseudo-labelled instances are taken into account. It verifies the effectiveness and advantageous of our CTV paradigm in exploiting unlabeled corpus for SRE tasks.

3.2.2 Effect of the amount of unlabeled data. To inspect the impact of the amount of unlabeled data on performance, we vary the ratio of unlabeled instances from the range of {0%, 10%, 30%, 50%, 70%} and fix the ratio of labelled instances to 10%. Experimental results are presented by Figure 2. We can observe that as the ratio of unlabeled instances increases, the F1-score first increases, then reaches the best and keeps stable afterwards, which reveals that most of unlabeled instances are helpful for labelled instances in constructing RE learners.

3.2.3 Quality of the pseudo-labelled instances. To check the quality of pseudo-labelled instances, we evaluate the accuracy of pseudo-labels based on their true labels. As shown in Figure 3, our CTV maintains competitive advantage both in accuracy and F1-score compared with other methods, which means the quality of pseudo-labelled data generated by CTV is higher than other methods. The reason for decline of accuracy is that easier instances will be pseudo-labelled in the early iterations, and the instances selected in the later iterations will be more difficult.

3.2.4 Impact of sampling ratio. To illustrate the effect of sampling ratio τ , we further conduct experiments by ranging it from

Table 2: Performance of CTV compared with baselines on Semeval.

	5%			10%			30%		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
PRNN	62.13±1.12	53.28±1.56	57.40±1.08	63.38±0.94	62.44±1.29	62.91±0.56	69.41±0.41	74.12±0.46	71.55±0.35
Self-training	59.50±0.78	61.18±0.75	60.29±0.63	63.07±0.57	67.75±0.64	65.34±0.38	68.72±0.52	75.41±0.68	71.84±0.41
DualRE	62.81±1.48	65.82±1.24	63.79±0.98	65.58±1.27	68.63±0.74	67.11±0.49	69.33±0.48	75.72±0.32	72.27±0.35
MrefG	63.36±1.57	65.18±1.02	64.24±0.97	66.44±0.82	70.81±0.94	68.60±0.65	70.30±0.51	74.81±0.73	72.45±0.44
Co-training	62.56±0.72	64.10±0.51	63.33±0.53	65.56±0.60	73.02±0.30	69.09±0.21	69.13±0.31	76.23±0.40	72.44±0.34
CTV	65.25 ±0.39	69.12±1.10	67.15±0.63	67.87±0.36	73.16±0.20	70.41±0.20	70.43±0.04	76.47±0.10	73.32±0.05
PRNN-gold	72.57±1.47	74.65±1.98	73.56±0.31	71.40±1.42	76.72±0.64	74.48±0.37	72.98±0.96	78.86±0.76	75.80±0.24

Table 3: Performance of CTV compared with baselines on Re-TACRED.

	3%			10%			15%		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
PRNN	64.31±1.78	58.89±1.39	61.47±1.04	70.17±1.64	66.89±1.39	68.47±0.12	73.81±0.83	69.39±0.46	71.52±0.63
Self-training	69.66±1.54	57.44±0.52	62.95±0.73	72.07±0.82	66.53±1.08	69.34±0.44	74.27±0.81	72.08±0.74	73.12±0.57
DualRE	70.55±0.51	59.20±0.36	64.37±0.32	73.73±1.08	68.96±0.87	71.26±0.38	75.52±0.91	73.42±0.53	74.45±0.18
MrefG	70.61±0.62	60.84±0.58	65.72±0.41	74.85±0.97	68.84±0.72	71.75±0.44	76.03±0.81	73.31±0.62	74.62±0.28
Co-training	72.05±0.94	61.05±0.84	66.16±0.15	74.48±0.83	68.77±0.47	71.51±0.56	76.58±0.90	72.53±0.50	74.49±0.18
CTV	73.92±0.29	65.04±0.71	69.19±0.32	75.08±0.55	70.65±0.80	72.79±0.20	78.61±0.54	73.28±0.49	75.85±0.07
PRNN-gold	77.38±1.96	78.24±0.72	77.78±0.63	78.93±1.26	76.74±0.46	77.84±0.39	79.47±0.35	77.27±0.33	78.35±0.13

Table 4: Effect of the hyper-parameter τ . The ratio of labelled and unlabeled instances are fixed to 10% and 50% respectively. The evaluation metric is F1-score (%).

Dataset \ τ	0.6	0.7	0.8	0.9	1.0
Semeval	68.71	69.86	70.22	70.41	70.17
Re-TACRED	69.95	71.18	72.37	72.79	72.33

Table 5: Performance of CTV with different base learners on Semeval with 5% labelled instances. ‘-CTV’ and ‘-gold’ represent our method and upper-bound respectively.

	PRNN	-CTV	-gold	R-BERT	-CTV	-gold
F1-score (%)	57.40	67.15	73.56	72.74	73.56	84.68

0.6 to 1.0. The results are presented in Table 4. As the sampling ratio increases, the F1-score first increases, then reaches the best, and decreases afterwards. The reasons are as follows: when τ is too small, some potential useful instances are discarded, especially in the minority relation class; however, when τ is too large, learners may encounter the instances with noise labels, which is harmful to the robustness of CTV.

3.2.5 Employing R-BERT as base learners. To verify the generality of CTV, we also evaluate its performance when R-BERT are employed as base learners. As shown in Table 5, CTV is capable of improving the RE performance with different base learners, which shows the generality of CTV for various learners.

4 RELATED WORK

Recently, several supervised methods for sentence-level RE have been proposed [13, 15, 17–20]. Despite enjoying great advantages, they face a tough challenge of labelled sentences scarcity. Thus,

Distant Supervision methods [8, 9, 14, 16] are proposed to leverage external knowledge bases to annotate more sentences. However, auxiliary knowledge bases do not exist in most of the domains.

Lately, Semi-supervised Relation Extraction (SRE) has achieved satisfying performance. Some early works are mainly based on self-training paradigm [10], which employs single learner to annotate some instances to refine itself iteratively. Such a paradigm is vulnerable to the issues of insufficient supervision and semantic drift. To tackle such issues, some works [4, 6] based on self-training with validation (STV) paradigm are proposed, which trains an external validator to assist the primal predictor in selecting pseudo-labelled instances with higher quality to retrain the learners.

Co-training [1] is another classic paradigm for semi-supervised learning. In the field of learning from noisy labels, a Co-training-like paradigm called Co-teaching [2] has shown its expressive power. Considering that Co-training and Co-teaching can mutually benefit, we integrate them in a unified framework in this paper.

5 CONCLUSION

Motivated by the fact that current SRE schemes under-utilize the potential of learners in exploiting unlabeled corpus, we proposed a novel CTV paradigm towards SRE tasks, which inherits the dual advantages of STV and co-training. In contrast to the popularly used STV paradigm for SRE tasks, CTV can benefit more from unlabeled corpus. Different from the regular co-training, CTV can generate more high-quality pseudo-labelled instances to achieve stronger generalization capability. Through extensive experiments carried out on two benchmarks, we demonstrated that our CTV consistently outperforms the state-of-the-art SRE schemes and works well with different types of RE learners. Besides RE, deploying CTV in other AI applications, where labelled data is scarce yet unlabeled data is readily available, is also well-worth studying in the future.

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